Chuanbo (Pilgrim) Peng

Tel: +1 (410)-916-0042 | Email: cpeng37@jh.edu | Baltimore, MD

EDUCATION

Johns Hopkins University, Baltimore, MD, USA — Aug 2026 – Expected Jun 2028

MSE in Applied Mathematics and Statistics

University of Nottingham, Nottingham, UK — Sep 2022 – Jul 2026

BSc (Hons) in Statistics | First Class Honours

RESEARCH EXPERIENCE

Selective Memory Reuse for LLM Agents — May 2026 – Aug 2026

Supervisor: Asst. Prof. Fang Kong, Southern University of Science and Technology

- Investigated selective retrieval and reuse of prior task trajectories for LLM agents on the AppWorld benchmark, focusing on when retrieved experience improves performance.
- Designed controlled experiments and abstention strategies to compare memory reuse against no-memory baselines and evaluate whether similarity and utility based signals reliably predict benefit.
- Conducted error and trajectory analyses to diagnose failure modes, identify limitations of proposed selection signals, and guide subsequent changes to the evaluation framework.

PROJECT EXPERIENCE

Numerical Modelling and Optimization of a 2D Thermostat-Heater Control System — Jan 2026 – Mar 2026

Applied mathematical and modelling project (Constrained PDE optimization), University of Nottingham

- Formulated a two-dimensional heat-diffusion model with Newton cooling boundary conditions and compared bang-bang and bounded-hysteresis control strategies.
- Defined a global objective functional measuring deviation from target temperature and used coarse-to-fine grid search to study thermostat placement and changes in the resulting optimization landscape.

Introduction to Multi-Armed Bandit Algorithms — Jul 2025 – Sep 2025

Instructor: Prof. Osman Yagan, Carnegie Mellon University

- Studied classical MAB algorithms including ETC, UCB, and Thompson Sampling, with emphasis on regret and sequential decision-making under uncertainty.
- Conducted an empirical study on Yelp and MovieLens comparing cumulative regret with inverse propensity scoring, using effective sample size to assess the reliability of offline evaluation.
- Authored a paper analyzing when IPS rankings agree with regret, finding that IPS can provide useful offline estimates but does not reliably preserve algorithm rankings or capture the dynamic cost of exploration.

SKILLS

Programming: Python (NumPy, SciPy, Pandas, Matplotlib), LaTeX

Machine Learning & Statistics: Regression, Bayesian Methods, Time Series

Optimisation: Convex Optimization, Gradient-Based Methods

Research: Experimental Design, Offline Evaluation, Statistical Modeling